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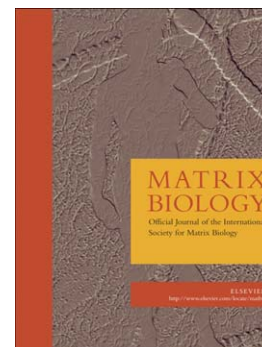
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Histidine-rich glycoprotein blocks collagen-binding integrins and adhesion of endothelial cells through low-affinity interaction with $\alpha 2$ integrin

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Abstract

The plasma protein histidine-rich glycoprotein (HRG) affects the morphology and function of both endothelial cells (ECs) and monocytes/macrophages in cancer. Here, we examined the mechanism of action of HRG's effect on ECs. HRG suppressed adhesion, spreading and migration of ECs specifically on collagen I (COL I) whereas ECs seeded on other extracellular matrix proteins were insensitive to HRG. HRG did not bind specifically to COL I or to the α -integrin binding site on collagen, GFOGER. Furthermore, HRG's inhibition of EC adhesion was not dependent upon heparan sulfate (HS) moieties as heparitinase-treated ECs remained sensitive to HRG. C2C12 cells expressing $\alpha 2$ integrin, the major collagen-binding α -integrin subunit in ECs, showed increased binding of HRG compared with wild type C2C12 cells lacking the $\alpha 2$ subunit. Recombinant $\alpha 2$ I-domain protein bound HRG and to a higher extent when in active conformation. However, the $\alpha 2$ I-domain bound weakly to HRG compared to COL I and the purified $\alpha 2\beta 1$ ectodomain complex failed to retain HRG. We conclude that HRG binds to $\alpha 2$ integrin through low-affinity interactions in a HS-independent manner, thereby blocking EC-adhesion to COL I.

Keywords:

HRG, adhesion, integrin, endothelial cells, collagen.

Abbreviations:

HRG – Histidine-rich Glycoprotein
HUVEC – Human Umbilical Cord Endothelial Cell
EC – Endothelial Cell
COL I – Type I Collagen
HS – Heparan Sulfate

1. Introduction

Histidine-rich glycoprotein (HRG) is a 73 kDa heparan sulfate (HS)-binding glycoprotein present in plasma at 150 µg/ml [1]. *Hrg* gene inactivation in mice is compatible with embryonic development and adult fertility, but gives rise to increased clot formation as well as increased fibrinolysis [2]. Reports from our and other laboratories strongly implicate mononuclear cells in HRG biology. HRG induces gene regulation changes in monocytes/macrophages [3] and iodinated HRG binds specifically to monocytic cells in a HS-independent manner [4].

Endothelial cells (ECs) and monocytic cells communicate in physiological and pathological processes e.g. by reciprocal exchange of growth modulatory factors [5, 6]. HRG and HRG-derived peptides have been shown to affect ECs, e.g. by suppressing endothelial chemotaxis towards growth factors [7, 8]. Moreover, ECs in the vasculature of HRG-expressing tumors attain a more mature morphology than non-HRG expressing tumors [9]. By binding and neutralization of proteins with anti-angiogenic properties such as thrombospondin and vasculastatin [10, 11], HRG may also indirectly, support angiogenesis.

The mode of action of HRG in affecting ECs has remained unclear. We therefore asked whether HRG acts directly on endothelial cells, and if so, through which mechanism. We show here that HRG suppressed EC attachment and spreading on type-I collagen (COL I), in a HS-independent manner. In contrast, EC attachment to other matrices was not affected by HRG. HRG bound to integrin $\alpha 2$ -expressing cells and bound with low efficiency to the integrin $\alpha 2$ protein I-domain, but not to the purified $\alpha 2\beta 1$ integrin ectodomain complex. We conclude that HRG's ability to interfere with EC adhesion and motility is critically dependent on the presence of $\alpha 2$ integrin, despite the lack of high-affinity interaction between the two proteins. The interplay may require other molecular partners yet to be identified. The data may have a bearing on the interpretation of HRG's effect on suppressing cancer growth and spread *in vivo*, which in part may depend on a primary effect of HRG on ECs.

2. Results

2.1 HRG negatively regulates endothelial cell migration and adhesion to COL I

HRG and peptides derived from HRG have previously been shown to inhibit endothelial cell migration *in vitro* [7, 8]. We sought to determine the underlying mechanism. First, we confirmed that with the current experimental set-up, purified, recombinant human HRG indeed inhibited migration of human umbilical vein ECs (HUVECs) through a COL I-coated membrane (Figure 1A). The effect of HRG was comparable to that of an integrin $\alpha 2$ -neutralizing antibody, which blocks adhesion of cells to COL I [12]. Moreover, HRG inhibited HUVEC adhesion to COL I-coated wells (Figure 1B).

To further explore HRG's modulation of HUVEC adhesion, we examined the potential matrix dependence in HRG's inhibition of HUVEC adhesion. As shown in Figure 1C, HRG inhibited HUVEC adhesion to COL I, but not to laminin, fibronectin or vitronectin. The specificity for HRG's inhibition of adhesion to COL I, but not to fibronectin, was recapitulated using the xCELLigence system, which measures current impedance changes following cell adherence and spreading on matrix (Figure 1D, E). The observation that HRG inhibits EC adhesion to COL I is particularly interesting as the localization of HRG in T241 fibrosarcomas tissue often occurred around collagen-rich vascular structures identified by

immunostaining for the vascular adhesion protein CD31 (Figure 2). Thus, HRG's COL I-specificity may have a significant *in vivo* role in modulating endothelial cell biology.

2.2 HRG does not directly alter HUVEC gene expression

We previously demonstrated that HRG affects expression of macrophage genes involved in the regulation of the extracellular matrix and immune responsiveness [3]. To examine whether a similar gene regulation pattern was observed in HRG-treated HUVECs, transcript arrays were performed. Of the 29,000 genes examined, no significant changes in gene expression were found when comparing HRG- and buffer-treated HUVECs at either 6 or 24 hours after addition of HRG at 1 μ g/ml (GEO Accession no GSE66768). This demonstrated that HRG's effects on HUVECs are not due to changes in gene expression over the time frames analyzed, consistent with the observation that HRG's inhibition of HUVEC adhesion can be seen in relatively short duration assays.

2.3 HRG does not bind specifically to COL I

We next sought to determine whether the specificity of HRG's inhibitory effects on EC adhesion and spreading on COL I was due to a specific binding of HRG to the matrix. We incubated various matrices with HRG and quantified bound HRG by ELISA [4]. While HRG inhibited cell adhesion only to COL I, it bound in a dose-dependent manner to all matrices tested compared to BSA-blocked wells (Figure 3A). However, the promiscuity of HRG's matrix interactions did not rule out the possibility that HRG might bind to different motifs in different matrices allowing specific interference with adhesion e.g. to COL I. To examine this possibility, we tested the ability of HRG to bind to the triple helical peptide GFOGER, the integrin-binding motif of COL I [12] (Figure 3B). HRG did however not display a specific interaction with GFOGER compared to the control peptides GPP and GPO, which have a triple-helix organization similar to that of GFOGER. Therefore, HRG's ability to inhibit endothelial cell adhesion to COL I was likely not related to matrix interactions. This suggests that HRG's inhibition of adhesion occurs on the integrin side of the integrin-matrix interaction, which is in line with our observation that pre-incubation of endothelial cells with HRG significantly enhanced adhesion blockade (Figure 3C). Moreover, when treating HUVECs with HRG followed by a wash-step, the inhibitory effect of HRG on HUVEC adhesion to COL I disappeared (Figure 3D). These results indicate HRG binds with low affinity to the EC surface and that HRG's effects on adhesion are not mediated via cell signalling changes.

2.4 HRG's inhibition of EC adhesion to COL I is independent of HS proteoglycan interactions

Binding to HS is known to facilitate and stabilize cell surface receptor-ligand interactions. Specifically, the HS-proteoglycan syndecan-1 supports high affinity binding of α 2 integrin to collagen. This mechanism seems to be independent of integrin conformational activation or direct binding of syndecan to collagen [13]. Consequently, loss of syndecan-1 results in a lower binding affinity and a reduction of α 2 β 1-mediated collagen adhesion. Given that HRG is a HS-binding protein [14], we investigated whether a potential interaction between cell-surface HS proteoglycans and HRG was responsible for HRG's effects on adhesion of ECs to COL I. We examined the ability of HUVECs to adhere to COL I and the GFOGER peptide after heparitinase treatment of cells, to enzymatically degrade cell surface HS moieties. As expected, heparitinase treatment of HUVECs suppressed adhesion and spreading of cells to COL I. This

effect was not observed for HUVEC adhesion to GFOGER, consistent with the redundant role for HS in $\alpha 2\beta 1$ integrin/GFOGER binding (Figure 4 A, B). Of note, heparitinase-treated HUVECs remained as sensitive to HRG as cells not treated with heparitinase. Moreover, the fact that HRG inhibited adhesion of HUVECs to GFOGER, which does not require $\alpha 2$ integrin-HS cooperation, agrees with the notion that HRG's inhibition of adhesion is independent of interference with HS-interactions.

2.5 HRG directly interacts with the COL I binding integrin $\alpha 2\beta 1$

We next chose to examine the potential interaction between HRG and the COL I-binding integrin $\alpha 2\beta 1$ [12], to provide an explanation for the inhibition of adhesion of endothelial cells to COL I. Integrin $\alpha 2\beta 1$ is the major COL I-binding receptor on HUVECs as determined by qPCR (Figure 5A). To explore if HRG treatment resulted in detectable changes in $\alpha 2\beta 1$ properties that might result in impaired integrin-COL I engagement, we examined the effect of HRG treatment on both $\alpha 2$ and $\beta 1$ surface expression, and $\beta 1$ activation status. HUVECs were treated with increasing concentrations of HRG or MnCl_2 , known to induce $\beta 1$ activation, followed by surface staining with either an antibody recognizing total $\beta 1$ or the activated $\beta 1$ subunit (Figure 5B). Neither changes in total surface $\beta 1$ expression (data not shown) nor relative changes in active $\beta 1$ integrin were observed. Treatment of HUVECs with HRG additionally caused no change in $\alpha 2$ cell surface expression (Figure 5C), suggesting that HRG did not mediate changes in the potential for either the $\alpha 2$ or $\beta 1$ integrins subunits to engage extracellular matrices. Finally, pull-down of the $\alpha 2$ subunit and subsequent blotting for $\beta 1$ integrin showed that HRG did not affect the association between the $\alpha 2$ and $\beta 1$ subunits (Figure 5D).

The lack of significant changes in the conformational activation or surface expression of integrins, led us to examine the potential binding of HRG to the intact cellular integrin complex where it may mediate its effect by steric hindrance of the integrin-matrix interaction. We examined the binding of HRG, fluorescently labelled with Alexa fluor[®] 555 dye (555-HRG), to mouse myoblast cell line C2C12 with and without $\alpha 2$ integrin expression. C2C12 cells expressing the $\alpha 2$ subunit (C2C12- $\alpha 2$) or wild type C2C12 cells (C2C12wt) were incubated with varying concentrations of 555-HRG for 1 hour at 4°C (to inhibit HRG internalization) followed by flow cytometry fluorescence analysis (Figure 6A). Although there was a basal level of HRG binding to $\alpha 2$ integrin-negative cells, likely due to the presence of cell surface HS proteoglycans, there was a significantly increased binding of HRG to $\alpha 2$ integrin expressing cells compared to wild type cells.

Next, we examined the possibility that HRG might bind to and mask the integrin I-domain, thereby inhibiting integrin $\alpha 2\beta 1$ -matrix interactions. For this purpose, we tested the ability of recombinant $\alpha 2$ I-domain as well as $\alpha 2$ I-domain containing the activation mutation E318W, to bind to HRG (Figure 6B) and assessed if HRG could inhibit binding of $\alpha 2$ I-domain protein to COL I-coated wells (Figure 6C). HRG binding to the $\alpha 2$ I-domain was weak but detectable and integrin activation increased the binding further (Figure 6B). Despite this however, HRG did not inhibit the interaction between COL I and the $\alpha 2$ I-domain (Figure 6C), nor did HRG bind to recombinant $\alpha 2\beta 1$ ectodomain protein (Figure 6D). These results suggest that HRG has multiple low affinity interactions with the $\alpha 2$ integrin on endothelial cells which specifically interferes with EC-COL I interactions.

3. Discussion

Here, we have elucidated a novel, COL I-specific, HS-independent and potentially regulatory role for HRG in vascular biology. HRG specifically suppressed EC adhesion to COL I and not to other matrix proteins. The introduction of $\alpha 2$ integrin into cells, lacking endogenous α -integrin subunit expression, lead to a considerable increase in HRG binding to the cells. While HRG bound weakly to the $\alpha 2$ I-domain and to an even higher degree to an activated mutant form of the $\alpha 2$ I-domain, we were unable to demonstrate significant HRG binding to purified $\alpha 2\beta 1$ protein. The $\alpha 2\beta 1$ -HRG interaction does not involve HS proteoglycans as HRG suppressed adhesion of ECs to COL I even after heparitinase digestion. Heparitinase-digested and undigested cells adhered equally well to the $\alpha 2$ -binding site on COL I, GFOGER, in keeping with the fact that $\alpha 2$ -binding to GFOGER does not involve HS [13]. Still, HRG inhibited adhesion of the heparitinase digested HUVECs to GFOGER.

Integrins are crucial in the regulation of endothelial cell and vessel biology. Tumor vessels are characterized by decreased stability, which in part is due to the decrease in basement membrane and pericyte coverage [15]. Consistent with the functional importance of integrin biology in pathological angiogenesis, a large number of pre-clinical and clinical trials are examining integrin blockade as cancer therapeutics [16], in particular, the collagen and fibronectin binding integrins [17].

We propose that the *in vitro* binding between HRG and $\alpha 2$ integrin adhesion complex is of such low affinity that it cannot be completely reconstituted with purified proteins. The high plasma levels of HRG (150 $\mu\text{g/ml}$) would indeed imply low affinity interactions *in vivo*. A high affinity interaction in the presence of such high systemic concentrations would result in dramatic deficits in EC-COL I interactions, likely not be consistent with the viability of *Hrg*^{-/-} mice [2]. HRG may additionally interfere with cell adhesion in a HS-dependent and therefore, promiscuous manner. In agreement, we previously showed that HRG-derived peptides bind to transformed endothelial cells and interfere with their matrix-interaction in a HS-dependent manner [14]. Moreover, binding of HRG to T lymphocytes and the consequent interference of cell-matrix interaction involves HS [18].

A range of endogenous angiogenesis inhibitors act by disrupting integrin-mediated adhesion and signalling of ECs in the cancer vasculature [17]. Integrins $\alpha \nu \beta 3$ and $\alpha \nu \beta 5$ are required for the effects of Canstatin, a type IV collagen fragment [19], and Endostatin, a collagen XVIII fragment [20]. Tumstatin, a type IV collagen fragment binds $\alpha \nu \beta 3$ and $\alpha 5 \beta 1$ [21] and Arresten, the 26-kDa non-collagenous domain of collagen IV, binds $\alpha 1 \beta 1$ [22]. Interestingly, several of these inhibitors are fragments of matrix proteins that are released from the tumor matrix by one or several of the proteases produced at elevated levels in cancers. The matrix fragments may exert their effects by blocking binding of endothelial integrins to the matrix; alternatively, fragments may bind to a site on the integrin distinct from the matrix binding site, and elicit specific, integrin-antagonistic signalling [23]. Of note, the exact details of the binding of the different inhibitors to their cognate integrin have not been characterized biochemically, most likely due to either the involvement of cofactors or to low affinity binding. It also cannot be excluded that HRG may modulate adhesion signalling through changes in the myristoylation status or other lipid modifications, important for proper signalling of membrane-associated signalling molecules including integrins. Indeed, a high throughput screen has previously identified tumor cell proliferation inhibitors that act in this manner [24]. At this point no such HRG-mediated effects upon lipid modifications have been described.

A number of reports strongly implicate HRG in the modulation of inflammation and inflammatory reactions. HRG regulates Fcγ receptor expression and macrophage phagocytosis [25]. HRG is moreover essential in mounting inflammatory and immune responses against bacterial and fungal infections [26, 27]. In cancer, HRG polarizes tumor-associated macrophages from a pro-angiogenic, immune-suppressive M2 phenotype towards an anti-tumor, immunity-promoting, M1 phenotype [9, 3]. In agreement, M2 polarization and cancer progression are enhanced in *hrg*^{-/-} mice [3]. We have previously shown that mouse tumours overexpressing HRG have altered vessel properties with more mature endothelial cells and better perfused vessels [9]. This change in vessel parameters in the presence of HRG could primarily be a result of HRG-induced polarization of tumor-associated macrophages from a pro-angiogenic M2 to a non-angiogenic M1. The current study suggests that HRG's interference with EC adhesion to COL I also could play an important role in the overall tumor vessel phenotype by suppressing neo-angiogenesis and endothelial migration. Additionally, the interaction of HRG and ECs may play a significant role in vascular homeostasis under physiological conditions.

4. Materials and Methods

4.1 Antibodies, Proteins, matrices and growth factors

The following antibodies were used: anti-HRG antibody (0119 antibody [8], the integrin-neutralizing antibody anti-α2 (Millipore, MAB 1950, for adhesion assays), anti-collagen type 1 antibody (AB765P, Millipore, for immunostaining), anti-mHRG (MAB1905, R&D), anti-CD31 (553370, BD Biosciences). Europium-labelled anti-GST antibody, Eu-labelled anti-His antibody and Eu-labelled anti-mouse-IgG antibody were purchased from PerkinElmer, Wellesley, MA, USA. Mouse monoclonal anti-human HRG antibody (MAB1869) was purchased from R&D Systems. Heparitinase (heparinase III) was obtained from Sigma Aldrich. For coating of plates for ELISA, adhesion assay or membranes for migration assay, collagen type 1 (#50053B, Advanced Biomatrix), fibronectin (F089535MG, Sigma Aldrich), vitronectin (V8379350UG, Sigma Aldrich), laminin (L202031MG, Sigma Aldrich), triple helical GPC-(GPP)5, GFOGER-(GPP)5-GPC-NH2 and control peptides (Auspep, Australia) were used. For production of recombinant human and mouse HRG (hHRG, mHRG), full-length HRG cDNA (hHRG; ENST00000232003 and mHRG; ENSMUST00000023590) including the signal sequence were cloned into the pCEP3Pu2 expression vector as described previously [8]. Human embryonic kidney HEK293-EBNA cells were transfected with the vectors. To avoid contamination with bovine HRG, the defined serum-replacement medium TCM (ICN Biomedicals) was used. HRG was purified using Ni-NTA agarose (Qiagen). Protein-containing fractions were pooled and dialyzed against TBS (pH 7.4). The protein preparation lacked lipopolysaccharide (endotoxin) contamination as determined using a sensitive chromogenic endotoxin quantification kit (Pierce). Recombinant α2I-domain was produced as a GST-fusion protein as described earlier [28]. Recombinant human integrin α2β1 ectodomain (5698-A2), mouse integrin α2β1 (7828-A2) ectodomain with His-tag was purchased from R&D Systems. Human recombinant VEGF-A165 was purchased from PeproTech.

4.2 Cell Culture and Animals

Human umbilical vein ECs (HUVECs; ATCC) were cultured on gelatin-coated dishes in Endothelial Cell Basal Medium MV2 (EBM32, C322221; PromoCell) with supplemental pack C339221, containing 5% foetal calf serum (FCS), epidermal growth factor (5 ng/ml), VEGF (0.5 ng/ml), fibroblast growth factor-2 (FGF2) (10 ng/ml), long R3 insulin growth factor-1 (20 ng/ml), hydrocortisone (0.2 μ l/ml), and ascorbic acid (1 μ l/ml). HUVECs at passage 3-6 were used. Wild type mouse C2C12 cells and integrin α 2 overexpressing C2C12 cells were cultured in DMEM 10% FCS, 2.5 μ l/ml puromycin for selection. For adhesion assays, ECs were used without serum starvation; however, C2C12 cells were serum starved overnight in DMEM with 1% FCS without growth factor supplement. For *in vivo* animal experiments female C57Bl/6 mice (8-10 weeks) were carried out in strict accordance with the ethical permit provided by the Committee on the Ethics of Animal Experiments of the University of Uppsala (Permit Numbers: C59/10, and C 224/10).

4.3 Gene expression array

The gene expression array was performed as described [3] on HUVECs treated with 500 ng/ml human recombinant HRG for 6 or 24h. The array data have been deposited with GEO (Accession no GSE66768).

4.4 Chemotaxis assay

Chemotaxis of HUVECs was assessed using a modified Boyden chamber, as described [8], on 8- μ m micropore polycarbonate filter (Nuclepore track-etched membrane, 155846, Whatman) coated with COL I solution at 160 μ g/ml. Cells were starved overnight with 1% FCS in MV2 medium, trypsinized and resuspended at 4.0×10^5 cells/ml in MV-2 (Promocell GmbH) 0.25% BSA, and Trasylol (Aprotinin, 495184, Bayer Healthcare) at 1,000 KIE/mL to inhibit trypsin. The cell suspension was added to the upper chamber and the chemoattractant, VEGF-A165, was added at 10 ng/ml to the lower chamber. HRG (500 ng/ml) was added to both chambers. After 5 hours at 37°C, cells that had migrated through the filter were stained with Giemsa and counted using the Image J software (<http://rsbweb.nih.gov/ij/>). All samples were tested at least in six wells per condition. Statistical significance was calculated using Students t test; $p < 0.05$ was considered significant.

4.5 Adhesion Assays:

4.5.1 Plate and Wash Assay. HUVECs and C2C12 cells were detached using non-enzymatic Cell Dissociation Solution (Sigma Aldrich), washed and resuspended in MV32 with 0.1% BSA. Cells were pre-incubated with HRG for 30 min at RT before application. Wells were coated with COL I (160 μ g/ml), fibronectin (50 μ g/ml), vitronectin (0.5 μ g/ml) or laminin 1 (20 μ g/ml) at 4°C overnight. After blocking plates with 10 mg/ml Pluronic acid (30085231, BASF) in PBS for 1 hour at 37°C, cells were added to wells and incubated at 37°C for 30 min. After gently washing 3 times with PBS, cells were quantified by either DAPI nuclear stain or using the CyQUANT Cell Proliferation Assay Kit (Invitrogen).

4.5.2 xCELLigence assay. Plates (96-well E-Plate 96; ACEA Biosciences, San Diego, CA, USA) were coated with GFOGER (3 μ g/cm²), COL I (5 μ g/cm²), fibronectin (5 μ g/cm²) or BSA (0.1%) overnight in 4°C. The wells were blocked for 1 hour at RT with 0.1% BSA. HUVECs were detached with 5 mM EDTA, washed once with PBS, incubated for 20 min at RT in serum-free DMEM supplemented with 5 mM MgCl₂ and

1000 ng/ml HRG, and seeded (15,000/well) to E-Plate 96. Cell adhesion and spreading was measured with the xCELLigence system (Roche Applied Science).

4.5.3 ELISA. NUNC MaxiSorp™ 96 well plates were coated with matrices in PBS; COL I (160 µg/ml), fibronectin (50 µg/ml), vitronectin (0.5 µg/ml) or laminin 1 (20 µg/ml), or incubated with PBS alone overnight at 4°C. Wells were washed with PBS containing 0.05% Tween 20 (PBST) before blocking with 10 mg/ml Pluronic acid in PBS for 1 hour at 37°C. Well were incubated with recombinant human HRG protein (0, 100, 500 ng/ml) for 2 hours at RT. After 3 washes with PBST, primary anti-HRG antibody (0119, 5 µg/ml) [8] was incubated for 1.5 hours at RT. After washing 5 times with PBST, secondary antibody (anti-rabbit IgG-HRP) was incubated at a 1:5000 dilution for 1 hour at RT. After 5 washes with PBST, a Peroxidase detection kit (KPL) was used and reaction was stopped with HCL. Absorbance was measured at 450 nm.

4.6 qPCR

Quantitative PCR was carried out using the TaqMan® qPCR system (Life Technologies). For quantitation the following predesigned TaqMan® probes/primer pairs were purchased from Life Technologies; Hs00235006_m1 (ITGA1, human α 1 Integrin), Hs00158127_m1 (human α 2 integrin), Hs0014623_m1 (human α 10 Integrin), Hs00201927_m1 (human α 11 Integrin), Hs00559595_m1 (human β 1 Integrin). Hs02800695_m1 (human HPRT1) was used for housekeeping. qPCR reactions were carried out using the TaqMan® Universal Master Mix II, as per manufacturers' instructions.

4.7 555-HRG binding assay

HRG was labelled with 555 dye using Alexa Fluor 555 Protein Labelling Kit (A20174, Invitrogen). C2C12 cells were dissociated with Cell Dissociation solution and suspended in DMEM with 0.5% BSA at a concentration of 1,000,000 cells/ml. After incubation with 0, 1 or 5 µg/ml 555-HRG on ice for 1 hour, cells were washed. Flow sorting analysis (FACS) was performed using FACS Calibur, and the data analysed with FACS DIVA 6.0 Software (BD Biosciences). Only intact cells were analysed, excluding cell debris. Students t test; $p < 0.05$ was considered significant.

4.8 Immunofluorescence

T241 mouse tumors were harvested from C57Bl/6 mice as described before [3]. Tumors were frozen in OCT compound, and 7 µm section cut on a cryostat. Slides were fixed with ice-cold methanol for 10 min, and antigen retrieval performed with Dako REAL Target Retrieval solution (S2031, Dako). Slides were blocked with 3% BSA, 5% FCS, 0.1% Triton X-100 in PBS for 1 hour at RT, and incubated with primary antibodies at 4°C overnight at a dilution of 1:400 (anti-mHRG), 1:200 (anti-collagen type 1) and 1:400 (anti-VE cadherin). After washing 3 times, secondary antibodies were incubated at 1:500 dilution for 1 hr at RT. After 5 washes with PBS, cells were stained with DAPI and the slides were mounted. Images were acquired on a Zeiss LSM710 confocal microscope.

4.9 Protein binding assays:

4.9.1 Integrin I-domain binding assay. Solid-phase binding assays were performed with a modification of a published method [28]. Plates (MaxiSorp® flat-bottom 96 well plate, Nunc) were coated overnight with

COL I (rat; 5 $\mu\text{g}/\text{cm}^2$) or HRG (human; 3 $\mu\text{g}/\text{cm}^2$) at 4°C. Wells were blocked for 1 hour at RT with BSA (PBS:Delfia Diluent II 1:1; PerkinElmer, Wellesley, MA, USA). After blocking, wells were washed once, and recombinant GST- α_2 I or activated mutant GST- α_2 IE318W (200 ng/well) was allowed to bind for 1 hour at RT in Delfia assay buffer (PerkinElmer, Wellesley, MA, USA) supplemented with 2 mM MgCl_2 . In the case of the HRG inhibition assay, 2000 ng/well recombinant GST- α_2 I was used and 0-720 ng/ml of HRG was added. After incubation, wells were washed 3 times and europium-labelled anti-GST antibody (PerkinElmer, Wellesley, MA, USA) added to the Delfia assay buffer. After 1 hour incubation, wells were washed 3 times, and label was dissociated with Delfia enhancement solution. Signal was detected with time-resolved fluorescence spectrophotometry (Victor3 multilabel counter; PerkinElmer).

4.9.2 Integrin ectodomain. For human integrin $\alpha_2\beta_1$ ectodomain, the assay was performed as described above but wells were coated with integrin (3 $\mu\text{g}/\text{cm}^2$) and bound HRG detected with mouse monoclonal anti-HRG-antibody (MAB1869, RnD Systems, USA) and Eu-labelled anti-mouse-IgG antibody (PerkinElmer, Wellesley, MA, USA).

4.9.3 HRG binding to collagen mimetic peptides. Assay was performed as described above but wells were coated with triple helical GFOGER/GPP/GPO (3 $\mu\text{g}/\text{cm}^2$) peptides and bound HRG was detected with mouse monoclonal anti-HRG-antibody (MAB1869) and Eu-labelled anti-mouse-IgG antibody.

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Author Contributions

FR, KS, SH, ST performed experimental work. FR, KS, LCW, JH and SJ were involved in experimental planning and manuscript preparation.

References

1. Lee C, B.E., Söllner C, Jahnen-Dechent W, Claesson-Welsh L, 2008. Structure and function of type 3 cystatins; fetuins, kininogen and histidine-rich glycoprotein. *Frontiers in Bioscience*.
2. Tsuchida-Straeten, N., Ensslen, S., Schafer, C., Woltje, M., Denecke, B., Moser, M., Graber, S., Wakabayashi, S., Koide, T., Jahnen-Dechent, W., 2005. Enhanced blood coagulation and fibrinolysis in mice lacking histidine-rich glycoprotein (HRG). *J Thromb Haemost* 3, 865-872.
3. Tugues, S., Honjo, S., König, C., Noguier, O., Hedlund, M., Botling, J., Deschoemaeker, S., Wenes, M., Rolny, C., Jahnen-Dechent, W., Mazzone, M., Claesson-Welsh, L., 2012. Genetic deficiency in plasma protein HRG enhances tumor growth and metastasis by exacerbating immune escape and vessel abnormalization. *Cancer Res* 72, 1953-1963.
4. Tugues, S., Roche, F., Noguier, O., Orlova, A., Bhoi, S., Padhan, N., Akerud, P., Honjo, S., Selvaraju, R.K., Mazzone, M., Tolmachev, V., Claesson-Welsh, L., 2014. Histidine-rich glycoprotein uptake and turnover is mediated by mononuclear phagocytes. *PLoS One* 9, e107483.
5. Eltzschig, H.K., Carmeliet, P., 2011. Hypoxia and inflammation. *N Engl J Med* 364, 656-665.
6. Hanahan, D., Weinberg, R.A., 2011. Hallmarks of cancer: the next generation. *Cell* 144, 646-674.
7. Dixelius, J., Olsson, A.K., Thulin, A., Lee, C., Johansson, I., Claesson-Welsh, L., 2006. Minimal active domain and mechanism of action of the angiogenesis inhibitor histidine-rich glycoprotein. *Cancer Res* 66, 2089-2097.
8. Olsson, A.K., Larsson, H., Dixelius, J., Johansson, I., Lee, C., Oellig, C., Bjork, I., Claesson-Welsh, L., 2004. A fragment of histidine-rich glycoprotein is a potent inhibitor of tumor vascularization. *Cancer Res* 64, 599-605.
9. Rolny, C., Mazzone, M., Tugues, S., Laoui, D., Johansson, I., Coulon, C., Squadrito, M.L., Segura, I., Li, X., Knevels, E., Costa, S., Vinckier, S., Dresselaer, T., Akerud, P., De Mol, M., Salomaki, H., Phillipson, M., Wyns, S., Larsson, E., Buysschaert, I., Botling, J., Himmelreich, U., Van Ginderachter, J.A., De Palma, M., Dewerchin, M., Claesson-Welsh, L., Carmeliet, P., 2011. HRG inhibits tumor growth and metastasis by inducing macrophage polarization and vessel normalization through downregulation of PlGF. *Cancer Cell* 19, 31-44.
10. Klenotic, P.A., Huang, P., Palomo, J., Kaur, B., Van Meir, E.G., Vogelbaum, M.A., Febbraio, M., Gladson, C.L., Silverstein, R.L., 2010. Histidine-rich glycoprotein modulates the anti-angiogenic effects of vasculostatin. *Am J Pathol* 176, 2039-2050.
11. Simantov, R., Febbraio, M., Silverstein, R.L., 2005. The antiangiogenic effect of thrombospondin-2 is mediated by CD36 and modulated by histidine-rich glycoprotein. *Matrix biology : journal of the International Society for Matrix Biology* 24, 27-34.
12. White, D.J., Puranen, S., Johnson, M.S., Heino, J., 2004. The collagen receptor subfamily of the integrins. *Int J Biochem Cell Biol* 36, 1405-1410.
13. Vuoriluoto, K., Jokinen, J., Kallio, K., Salmivirta, M., Heino, J., Ivaska, J., 2008. Syndecan-1 supports integrin $\alpha 2 \beta 1$ -mediated adhesion to collagen. *Exp Cell Res* 314, 3369-3381.
14. Vanwildemeersch, M., Olsson, A.K., Gottfridsson, E., Claesson-Welsh, L., Lindahl, U., Spillmann, D., 2006. The anti-angiogenic His/Pro-rich fragment of histidine-rich glycoprotein binds to endothelial cell heparan sulfate in a Zn^{2+} -dependent manner. *J Biol Chem* 281, 10298-10304.
15. Baluk, P., Morikawa, S., Haskell, A., Mancuso, M., McDonald, D.M., 2003. Abnormalities of basement membrane on blood vessels and endothelial sprouts in tumors. *Am J Pathol* 163, 1801-1815.
16. Weis, S.M., Cheresh, D.A., 2011. αV integrins in angiogenesis and cancer. *Cold Spring Harbor perspectives in medicine* 1, a006478.
17. Goodman, S.L., Picard, M., 2012. Integrins as therapeutic targets. *Trends in pharmacological sciences* 33, 405-412.
18. Olsen, H.M., Parish, C.R., Altin, J.G., 1996. Histidine-rich glycoprotein binding to T-cell lines and its effect on T-cell substratum adhesion is strongly potentiated by zinc. *Immunology* 88, 198-206.

19. Magnon, C., Galaup, A., Mullan, B., Rouffiac, V., Bouquet, C., Bidart, J.M., Griscelli, F., Opolon, P., Perricaudet, M., 2005. Canstatin acts on endothelial and tumor cells via mitochondrial damage initiated through interaction with $\alpha v \beta 3$ and $\alpha v \beta 5$ integrins. *Cancer Res* 65, 4353-4361.
20. Rehn, M., Veikkola, T., Kukk-Valdre, E., Nakamura, H., Ilmonen, M., Lombardo, C., Pihlajaniemi, T., Alitalo, K., Vuori, K., 2001. Interaction of endostatin with integrins implicated in angiogenesis. *Proc Natl Acad Sci U S A* 98, 1024-1029.
21. Sudhakar, A., Sugimoto, H., Yang, C., Lively, J., Zeisberg, M., Kalluri, R., 2003. Human tumstatin and human endostatin exhibit distinct antiangiogenic activities mediated by $\alpha v \beta 3$ and $\alpha 5 \beta 1$ integrins. *Proc Natl Acad Sci U S A* 100, 4766-4771.
22. Nyberg, P., Xie, L., Sugimoto, H., Colorado, P., Sund, M., Holthaus, K., Sudhakar, A., Salo, T., Kalluri, R., 2008. Characterization of the anti-angiogenic properties of arresten, an $\alpha 1 \beta 1$ integrin-dependent collagen-derived tumor suppressor. *Exp Cell Res* 314, 3292-3305.
23. Nyberg, P., Xie, L., Kalluri, R., 2005. Endogenous inhibitors of angiogenesis. *Cancer Res* 65, 3967-3979.
24. Bhandarkar, S.S., Bromberg, J., Carrillo, C., Selvakumar, P., Sharma, R.K., Perry, B.N., Govindarajan, B., Fried, L., Sohn, A., Reddy, K., Arbiser, J.L., 2008. Tris (dibenzylideneacetone) dipalladium, a N-myristoyltransferase-1 inhibitor, is effective against melanoma growth in vitro and in vivo. *Clin Cancer Res* 14, 5743-5748.
25. Chang, N.S., Leu, R.W., Rummage, J.A., Anderson, J.K., Mole, J.E., 1992. Regulation of macrophage Fc receptor expression and phagocytosis by histidine-rich glycoprotein. *Immunology* 77, 532-538.
26. Rydengard, V., Shannon, O., Lundqvist, K., Kacprzyk, L., Chalupka, A., Olsson, A.K., Morgelin, M., Jahnen-Dechent, W., Malmsten, M., Schmidtchen, A., 2008. Histidine-rich glycoprotein protects from systemic *Candida* infection. *PLoS pathogens* 4, e1000116.
27. Shannon, O., Rydengard, V., Schmidtchen, A., Morgelin, M., Alm, P., Sorensen, O.E., Bjorck, L., 2010. Histidine-rich glycoprotein promotes bacterial entrapment in clots and decreases mortality in a mouse model of sepsis. *Blood* 116, 2365-2372.
28. Zhang, W.M., Kapyla, J., Puranen, J.S., Knight, C.G., Tiger, C.F., Pentikainen, O.T., Johnson, M.S., Farndale, R.W., Heino, J., Gullberg, D., 2003. $\alpha 11 \beta 1$ integrin recognizes the GFOGER sequence in interstitial collagens. *J Biol Chem* 278, 7270-7277.

Figure Legends:**Figure 1.**

HRG suppresses EC-COL I interactions. (A) Migration in a modified Boyden chamber of HUVECs toward VEGF-A165 (10 ng/ml), in the absence and presence of HRG. (B) Adhesion of HUVECs to COL I matrix with and without HRG. (C) Adhesion of HUVECs to COL I, fibronectin (FN), vitronectin (VN), laminin 1 (Lam) or bovine serum albumin (BSA) with and without HRG. (D and E) Adhesion and spreading of HUVECs to COL I (D) and fibronectin (E) matrices with and without HRG as indicated. Recombinant human HRG was given at 500 ng/ml in all panels. Detailed conditions are given in Methods. Error bars represent SEM. * $p < 0.05$, Student's t-test.

Figure 2.

In vivo localization of HRG. HRG demonstrated deposition around collagen-rich vascular structures in T241 fibrosarcoma tumors in vivo. Left - DAPI (grey), CD31 (red), collagen (yellow). Right – DAPI (grey), HRG (Rainbow RGB false colour from blue; low expression to red; high expression). Arrows identify collagen-rich vessels localizing with abundant HRG deposition.

Figure 3.

HRG binding assays. (A) HRG's binding to various matrices, assessed by ELISA. (B) Binding of HRG to the $\alpha 2$ integrin-binding site on collagen, GFOGER, to the control peptides GPP and GPO, and to BSA, assessed by time-resolved fluorescence spectrophotometry. (C) Enhanced inhibition by pre-incubation of ECs with HRG for 30 min compared with mixing HRG with ECs at the onset of adhesion to COL I (0 min). (D) HUVEC adhesion. HUVECs were treated with and without HRG followed by a wash-step as indicated (+wash). Cells were then seeded on tissue culture plastic coated with COL I, and HUVEC adhesion was monitored. HUVEC adhesion was inhibited by HRG only in the unwashed condition. Adhesion is expressed normalised to untreated cells. Error bar represent SEM. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.0001$; Student's t-test.

Figure 4.

HRG's inhibition of adhesion is HS-independent. (A) HUVEC adhesion and spreading to collagen. (B) HUVEC adhesion and spreading to GFOGER peptide. HUVECs were treated or not with heparitinase (Htase) as indicated. Figures represent 2 experimental replicates; error bars represent SEM.

Figure 5.

HRG does not interfere with $\beta 1$ integrin activation or complex formation with $\alpha 2$. (A) qPCR analysis of α integrin subunit expression in HUVECs. (B) and (C) integrin subunit analysis. Cells were treated with increasing concentrations of HRG as indicated and analyzed for changes in $\beta 1$ activation status (B) or in total surface $\alpha 2$ expression (C). (D) $\alpha 2$ - $\beta 1$ complex formation. $MnCl_2$ indicates treatment of cells with $MnCl_2$ buffer to activate integrins, as a positive control. HUVECs were treated with HRG and $\alpha 2$ integrin was immunoprecipitated followed by blotting for $\beta 1$ integrin. Values are expressed as the ratio of $\beta 1$ to $\alpha 2$ integrin. Error bars represent SEM.

Figure 6.

Figure 6. Binding of HRG to $\alpha 2$ integrin on cells. (A) Binding of 555-labelled HRG to C2C12 wild type (wt) and $\alpha 2$ integrin-expressing ($\alpha 2$) C2C12 cells. (B) Binding of wt or activating mutant $\alpha 2$ I- domain to HRG- BSA- or COL I-coated wells. (C) Lack of HRG inhibition of the interaction between $\alpha 2$ I-domain and COL I. (D) Lack of HRG binding

to recombinant $\alpha 2 \beta 1$ integrin. Wells were coated with $\alpha 2$ integrin I-Domain, BSA or HRG (positive control), incubated with HRG and probed with an anti-HRG antibody followed by a europium-labelled secondary antibody. Time resolved fluorescent spectroscopy was used in B-D. Error bars represent SEM. * $p < 0.05$, ** $p < 0.01$. Student's t-test.

Figure 1. Roche et al.

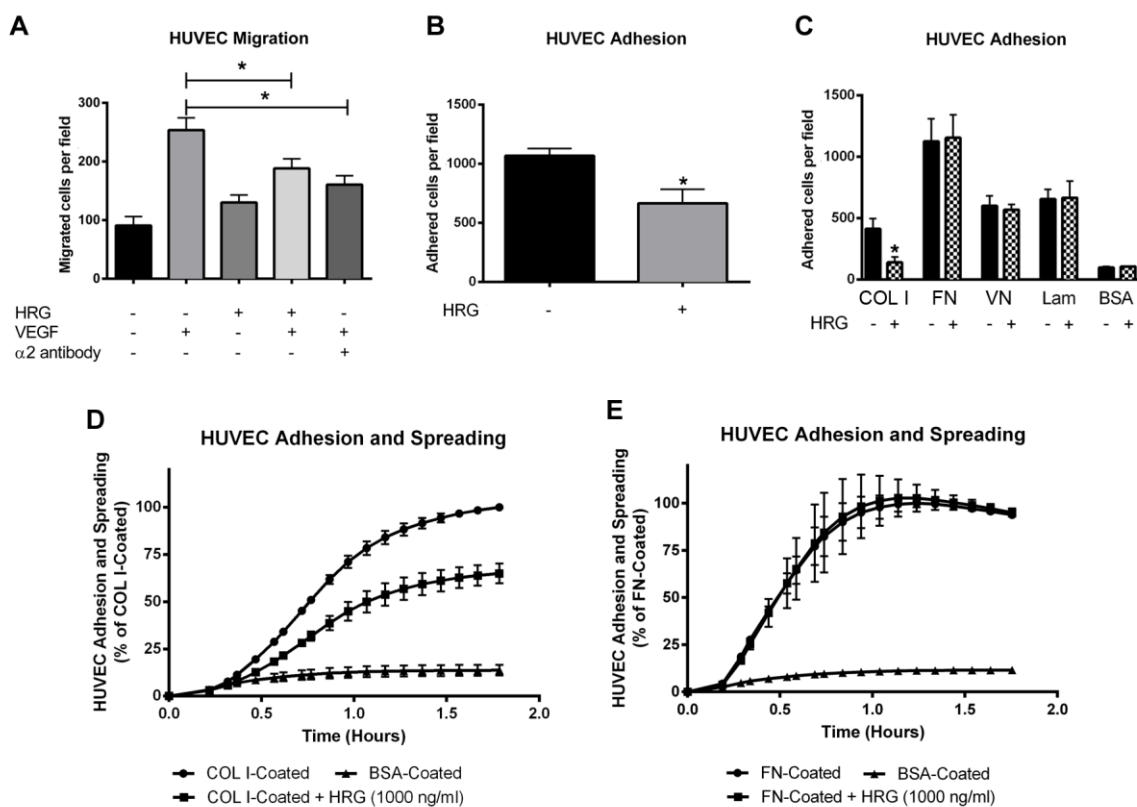


Figure 2. Roche et al.

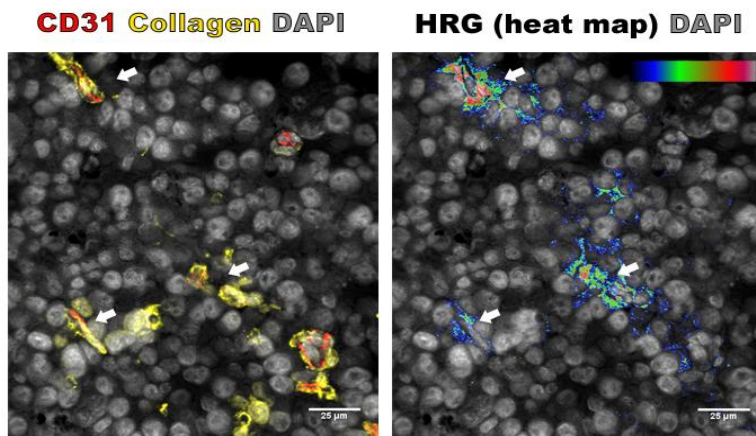


Figure 3. Roche et al.

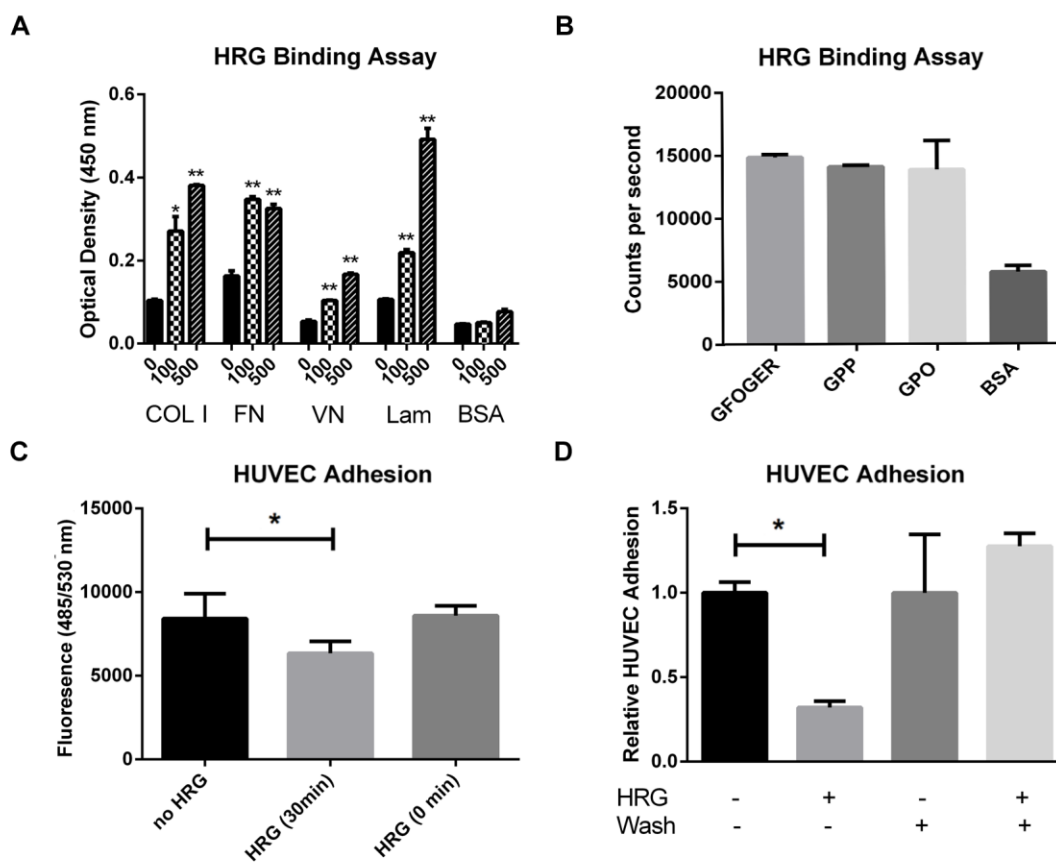


Figure 4. Roche et al.

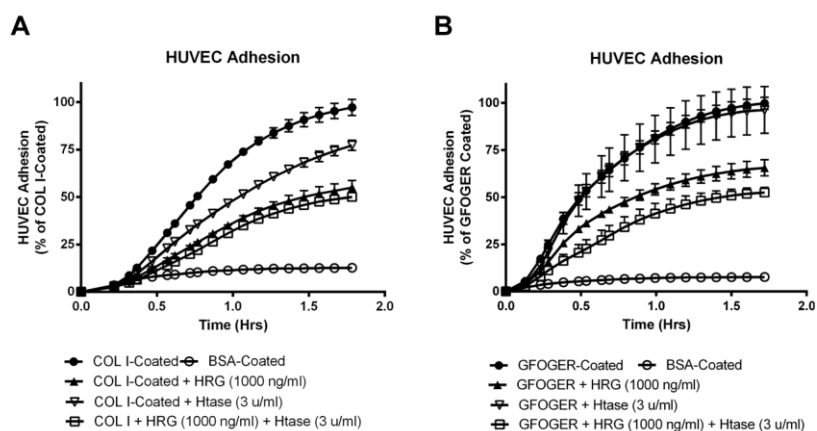


Figure 5. Roche et al.

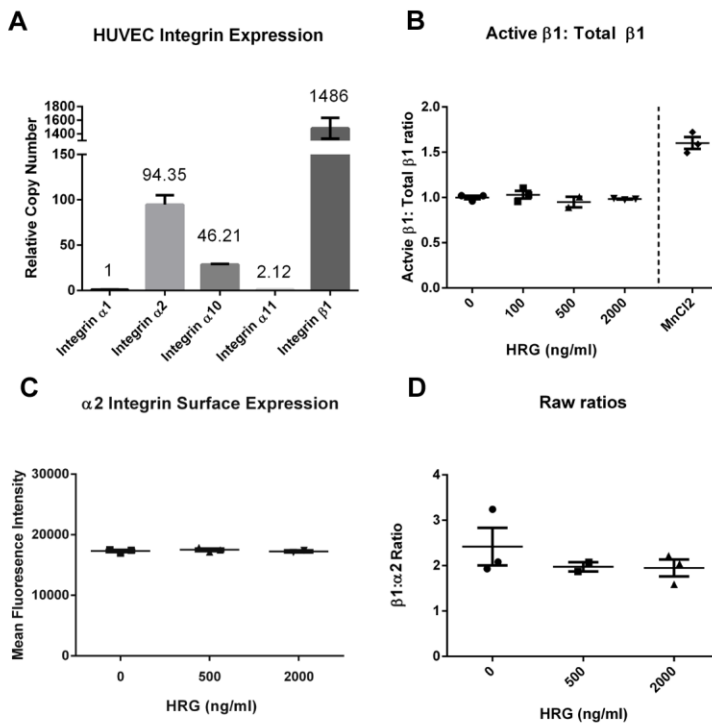
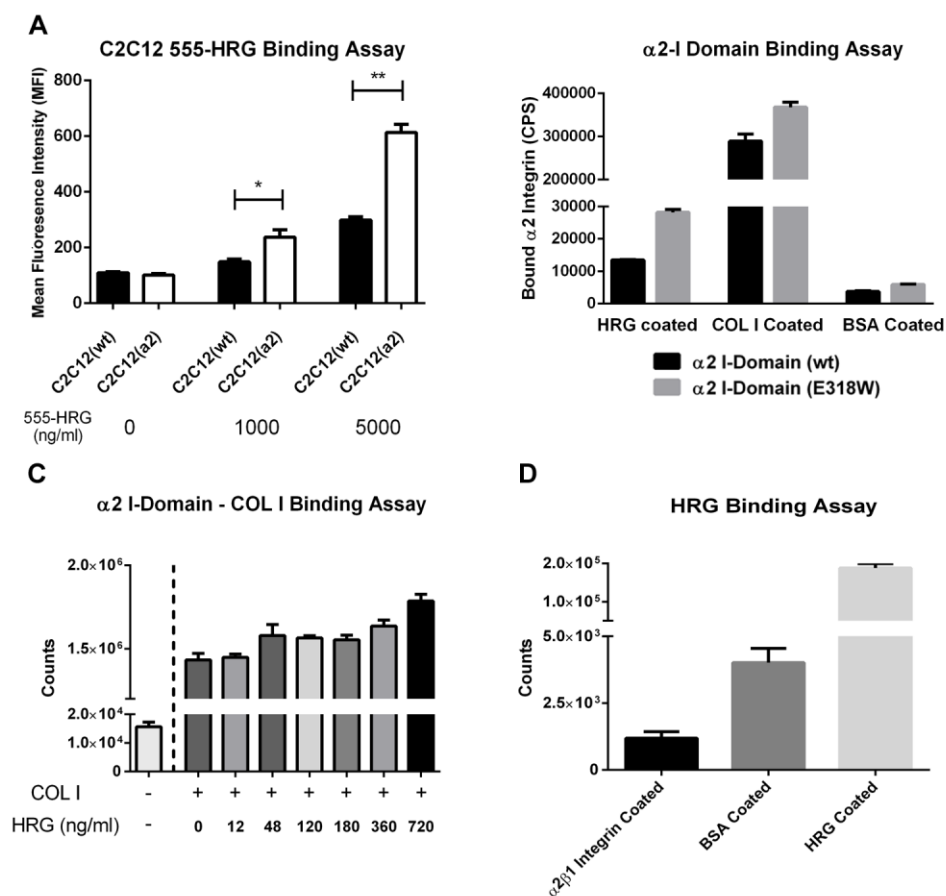


Figure 6. Roche et al.



Highlights:

The plasma protein HRG inhibits endothelial cell adhesion, spreading and motility.

HRG inhibits endothelial cell adhesion specifically to collagen I.

HRG inhibits adhesion through low affinity interaction with $\alpha 2$ integrin, independent of heparan sulfate.

HRG colocalizes with collagen and vessel-rich stroma in tumors.